TO 33kV SWITCHBOARD F00-SW-010 PANEL B04
REFER 2000-F00-SLD-EL-10002

300/1A 2.5VA 5P20

F28-TR-013
33 /11 kV
12500 kVA
Dyn11
z=8%
KNAN

F28-NR-013
ULTRAFINE GRINDING 2/3 TRANSFORMER
F28-TR-013 NEUTRAL EARTHING CUBICLE

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

INTERTRIPS TO SIGNALS

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

NOTE 4

F28-SR-001
ULTRAFINE GRINDING 11kV SWITCHROOM

E-SHUT F28SR001-LCS
NOTE 5

F28-SW-013
ULTRAFINE GRINDING 2/3 11kV SWITCHBOARD

TEST TEST

900/1A 2.5VA Cl 0.2 — PQM SEL735
900/1A 2.5VA 5P20 — 50 51 PR NOTE 2 SEL 751 AFD
900/1A 2.5VA 5P20
PR 50G SE-330AU
EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS
FAULT
CLOSE TRIP 1250A
87T
26Q 49W 63 71 PR SEL 787

2000-F28-SCM-EL-10011 A01 NOTE 2
2000-F28-SCM-EL-10103 A02
2000-F28-SCM-EL-10107 A04
2000-F28-SCM-EL-10019 A05
11kV 1250A 3-E 25kA FOR 3s

2A 630A NOTE 2
VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

NOTE 3
A03 2000-F28-SCM-EL-10015
CLOSE TRIP 630A
50 50G 27 AFD 46 38 51 66 49 49R PR SEL 710-5
300/1A 2.5VA 5P20 TEST
50/1A 1.5VA 10P10 TEST

NOTE 3
CLOSE TRIP 630A
50 50G 27 AFD 38 51 51G 81 66 49R PR SEL 710
300/1A 2.5VA 5P20 TEST
50/1A 1.5VA 10P10 TEST

NOTE 3
CLOSE TRIP 630A
50 50G 27 AFD 51 51G 81 PR SEL 751
250/1A 2.5VA 5P20 TEST
50/1A 1.5VA 10P10 TEST

UFG MILL PACKAGE
SEL 2600
MILL MOTOR
RTD
M 3~
3800 kW
LRS I/O TO DCS RIO PANEL
F28-MUF-402
ULTRAFINE GRINDING MILL 2
2000-F28-SCM-EL-10013

UFG MILL PACKAGE
SEL 2600
MILL MOTOR
RTD
M 3~
3800 kW
LRS I/O TO DCS RIO PANEL
F28-MUF-403
ULTRAFINE GRINDING MILL 3
2000-F28-SCM-EL-10017

11kV SPARE
FULLY EQUIPPED
-

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CB COMPARTMENT IS NOT CURRENT SUPERVISED. TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 33kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
26Q	TRANSFORMER PROTECTION DEVICES
27	UNDER VOLTAGE
38	BEARING RTD TEMPERATURE
46	CURRENT IMBALANCE
49	THERMAL OVERLOAD
49R	RTD BIASED THERMAL OVERLOAD
50	INSTANTANEOUS OVERCURRENT
50SG	INSTANTANEOUS OVERCURRENT SENSITIVE GROUND
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
63	TRANSFORMER PRESSURE / BUCCHOLZ
64	EARTH FAULT
66	NOTCHING OR JOGGING DEVICE / MAXIMUM STARTS PER HOUR / TIME BETWEEN STARTS
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
87T	TRANSFORMER DIFFERENTIAL
AFD	ARC FLASH DETECTION

2000-F28-SLD-EL-10001_Rev2

Single line diagram

D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F28 - ULTRAFINE GRINDING 2 & 3\2000-F28-SLD-EL-10001.DWG 29SEP26 7:25AM ROBIN.LAW